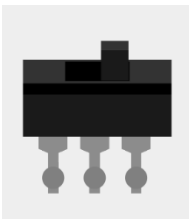
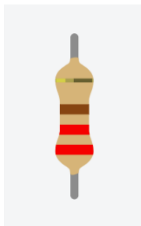
# Fortune Machine Project

For this project, you'll use a **few types of Switches** as well as a **Liquid Crystal Display** to create a *fortune generating machine.* The concept of pull-up resistors in the context of reading switch states will be reinforced as well.

## Materials:

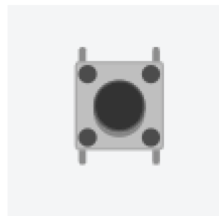


**1 x** Slide Switch



Resistor:
**3 x** 220Ω


**1 x** Tilt Switch

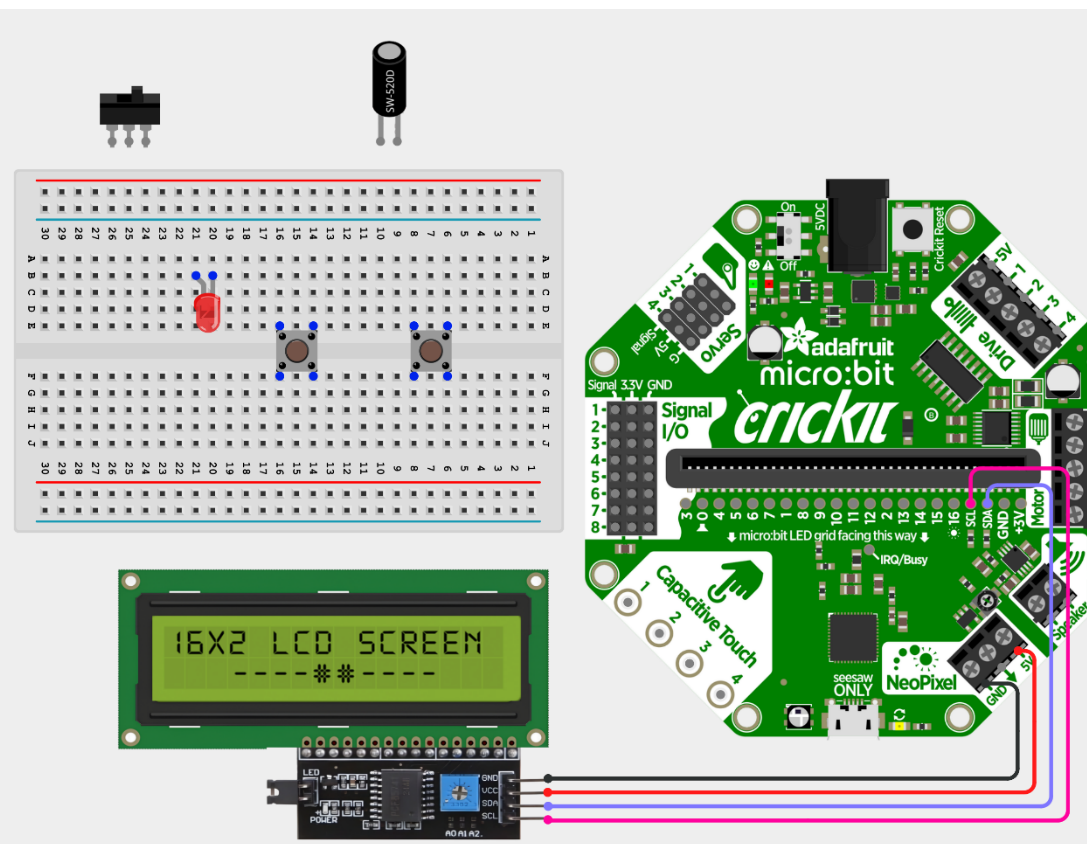


**2 x** Pushbuttons


**1 x** 1602 LCD w/ i2c adapter

## Wiring Diagram Version 1: (Challenge Yourself)

Below is a partial wiring diagram. Try to complete the rest of the circuit on your own, according to the connections chart on the right. *A complete diagram is available on the next page to check your work against.*




**Component Connections:**

<u>Tilt Switch:</u>

- Terminal 1 → pin 15
- Terminal 2 → GND

** For this circuit enable the internal pull-up resistor on P15.*

<u>Push Button (left)</u>

- Terminal 1 → Crickit signal 2
- Terminal 2 → GND

<u>Push Button (right)</u>

- Terminal 1 → Crickit signal 3
- Terminal 2 → GND

** Both pushbuttons should be pulled high with an external pull-up resistor.*

<u>Slide Switch</u>

*Slide switches don't typically fit directly into a breadboard; we'll use a switch with wires soldered to it to connect it to the circuit.*

- Common (center) → GND
- Left Terminal → pin 12
- Right Terminal → none

** For this circuit enable the internal pull-up resistor on P12.*

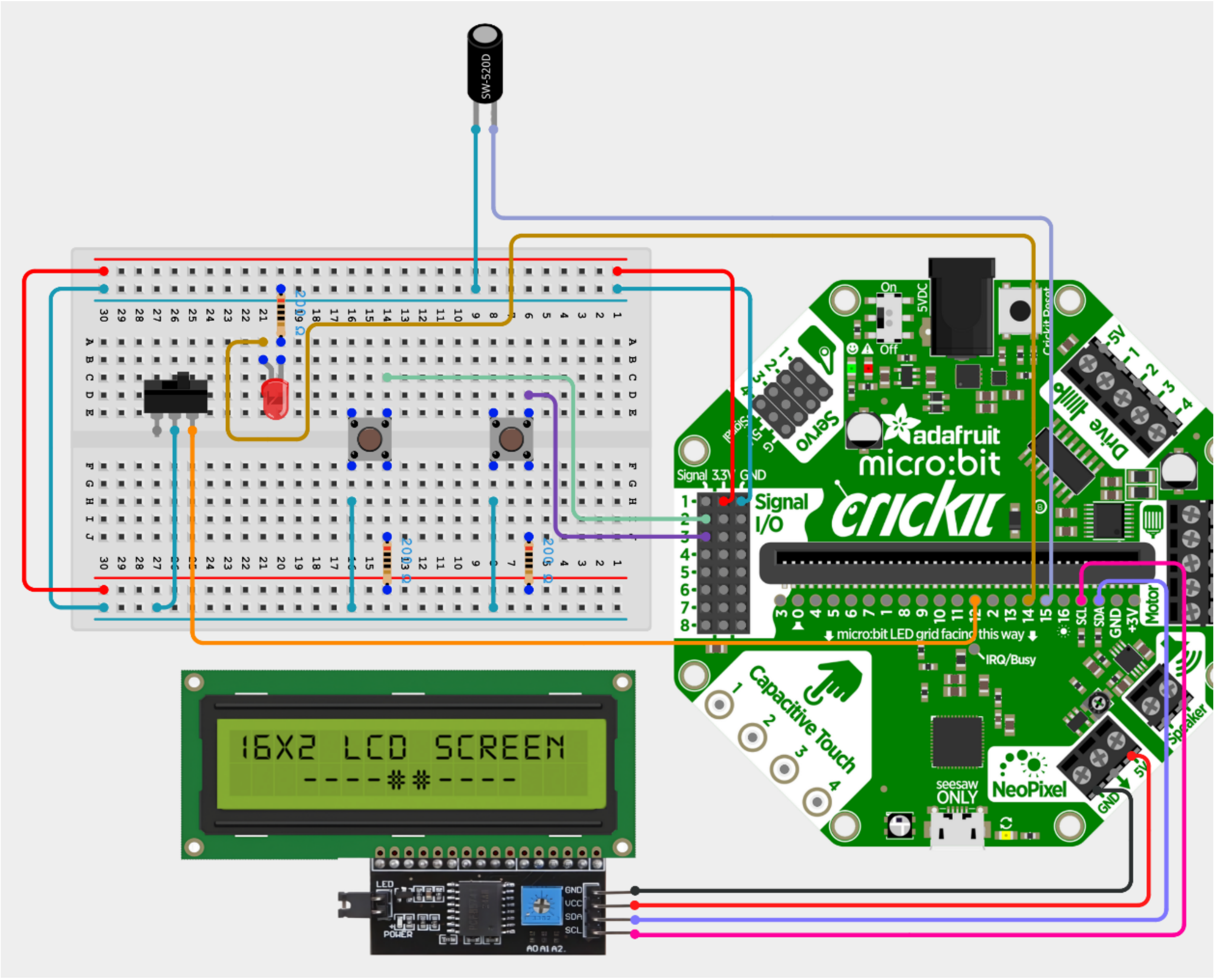
<u>LED</u>

- Cathode → resistor to GND
- Anode → pin 14

** Recall that Anode is the long lead (+)*

# Wiring Diagram Version 2: (Check Your Work)

Cross-reference your work with the full wiring diagram below!



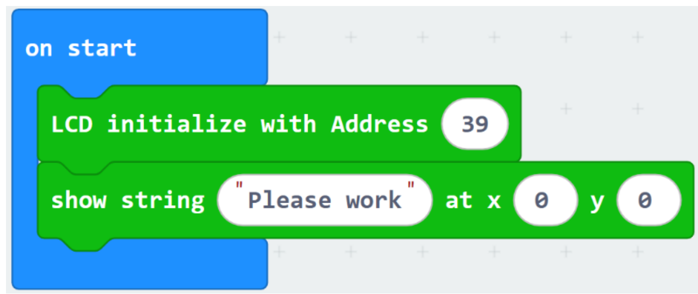
## Verify Circuit

As the complexity of this circuit is increasing, so is the possibility of errors in component placement (or poor connections for the physical circuit). Before getting into the complexity of the whole project, it would be a good idea to **verify that you've wired the components correctly**, by individually running a simple test on each.

**Note:** *Once you're done a test, disable that code and start on the next test. You shouldn't need to run all the tests concurrently.*

## Test One: Liquid Crystal Display

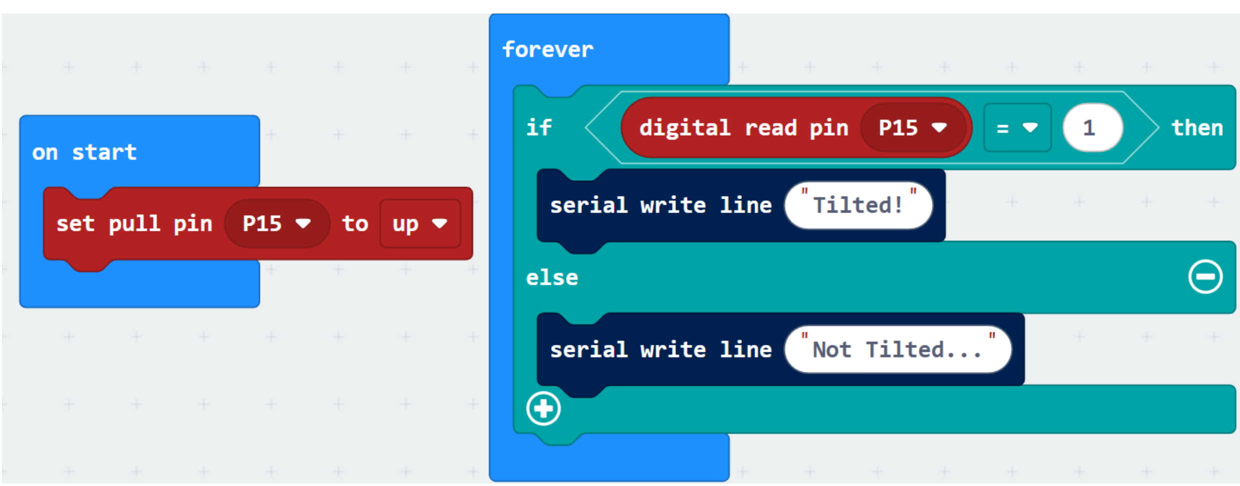
There are a several *Extensions* that allow us to communicate with the LCD. For this project, we need to use one that allows for scrolling of text. Add the extension shown below and try the provided test blocks to ensure the component is working correctly.


*Recall that you might need to adjust the potentiometer on the back of the module to improve the contrast if the text is difficult to see.*

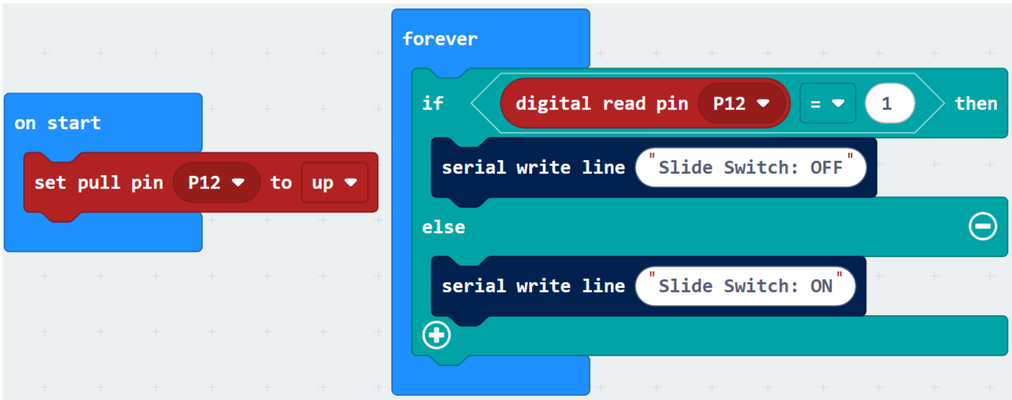
## Test Two: Tilt Switch

The tilt switch should be using the *internal pull-up resistor*, so don't forget to enable the resistor through code.



## Test Three: Slide Switch

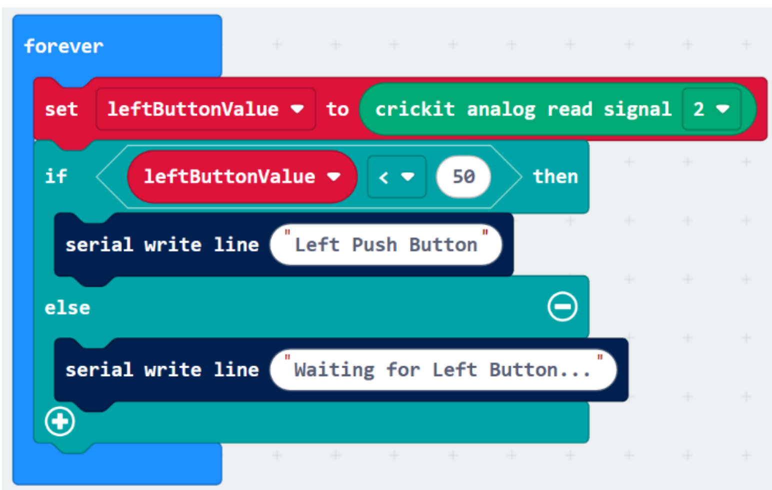
This test should be very similar to the tilt switch, as both have the same type of circuit.




## Test Four: Pushbuttons

Since the pushbuttons are to be read using the *Crickit Signal Pins*, recall that you'll need to use **analogRead.**

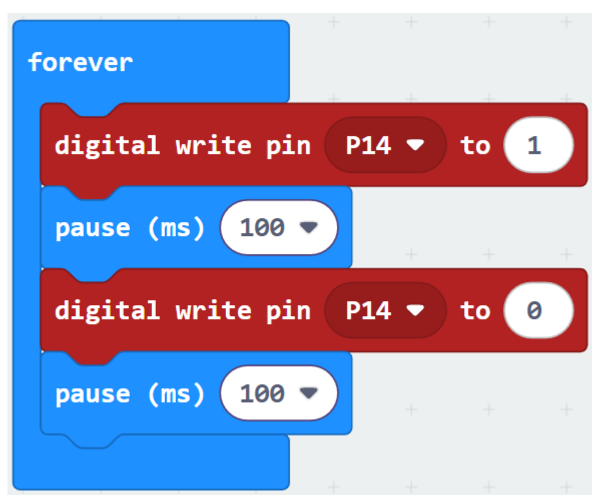
- When the button isn't pressed, the reading should be high (above 1000).
- When the button is pressed, the reading should be low (around 2-3)




For the right pushbutton you can do the same, just create a **rightButtonValue** variable and change the analogRead to check **Crickit Signal 3**.

## Test Five: LED

Easy to test, just cycle it OFF and ON using a loop…

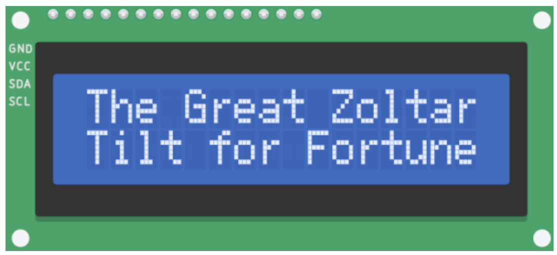


## Assignment Details: Base Features

For this assignment, completing the following base features will earn you a maximum mark of 10/10. Some hints and explanation will be given, but you'll need to construct a code solution on your own.
*It may be very useful to look back at your previous demos as reference material.*

Tell Them What to do:

The LCD Module will be the main way to communicate with the user. When the program first starts, have the module display a title and basic instructions, something like this:




Tilt For Fortune Five:

Your program should have at **five possible fortunes** which are chosen from randomly each time a tilt event occurs. Here are a couple details about these fortunes:

- Before showing the random fortune, there should be a **3 – 2 – 1 countdown** shown on the LCD.
- Once the countdown is done being displayed, show randomly chosen fortune on the LCD module.
  - *The chosen fortune should stay visible until the next time the tilt switch is activated.*
- You can choose what your possible fortunes will be, but one of them must be along the lines of "Today you are going to have great luck!"

*One purpose of the countdown (beyond building suspense) is to allow lots of time for the user to un-tilt the switch before the fortune is shown. This should reduce the likelihood of one tilt action resulting in multiple fortunes being drawn.*

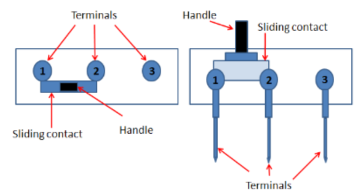
Scrollability:

Longer fortunes can't be displayed on the LCD all at once. Let's work around this limitation by adding user inputs to scroll the text being shown on the display:

- When the **left** pushbutton is pressed, scroll the LCD's text to the left.
- When the **right** pushbutton is pressed, scroll the LCD's text to the right.

Always Lucky:

The final feature for this project is to add a bit of a cheat to the fortune machine: a switch that can **guarantee a good fortune**.




- IF the switch is in the ON position:
  - The user always gets the best fortune
  - LED is lit up to show that always lucky mode is **enabled**.
- IF the switch is in the OFF position:
  - Fortunes are randomly selected from all the available options
  - LED is off to show that always lucky mode is **disabled**.

**------ Suggestions and Hints are below but try to build what you can on your own first! ------**

## Hints and Suggestions:

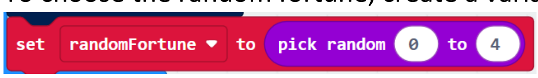
<u>Tell Them What to Do:</u>

This feature is straightforward. Just be sure to

- Initialize the LCD module
- Display top line "The Great Zoltar"
- Move the cursor to **row 1**, **col 0**
- Display the bottom line "Tilt For Fortune"

<u>Tilt For Fortune Five:</u>

- Start from detecting a tilt event. Look back at our demos if needed.
  - Since this switch is using an *internal pull-up resistor,* a **digitalRead()** value of **1** should occur <u>when the switch is tilted.</u>
- For the countdown, you can either slowly write all the text to be on the screen at once
  - Use the `pause (ms)` block to control wait times between text being displayed.
- To choose the random fortune, create a variable and choose a random (0 to 4) number in it:



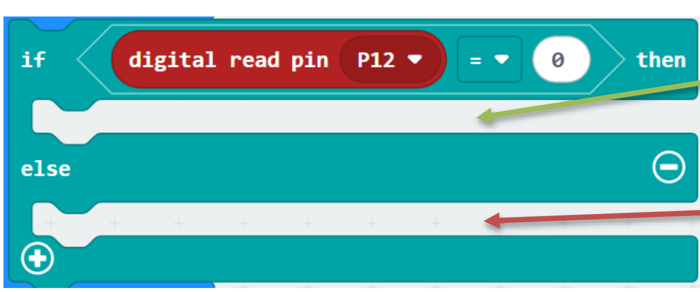

  - Each random number should represent <u>one of the five possible fortunes.</u>
  - Follow this up with an **IF, ELSE IF, ELSE IF, ELSE IF, ELSE** *(5 branches)*, one option for displaying each of the <u>5 possible values of **randomFortune.**</u>


<u>Scrollability:</u>

- Ensure you can reliably read the buttons (see notes in *Hardware Tests* above).
- Then just have each button cause either `Shift Left` or `Shift Right` to run.

<u>Always Lucky:</u>

- Again, make sure you first can read the switch position reliably, then check for the "ON" position




Means switch is ON:
- Turn on LED
- Display Lucky Fortune

Means switch is OFF:
- Turn off LED
- Put your regular "pick random fortune out of five options" code here…